Every force has a source and a receiver

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Receiver / ES: objeto que recibe la fuerza / PT: objeto que recebe a força

The object a force acts on; also name its source.

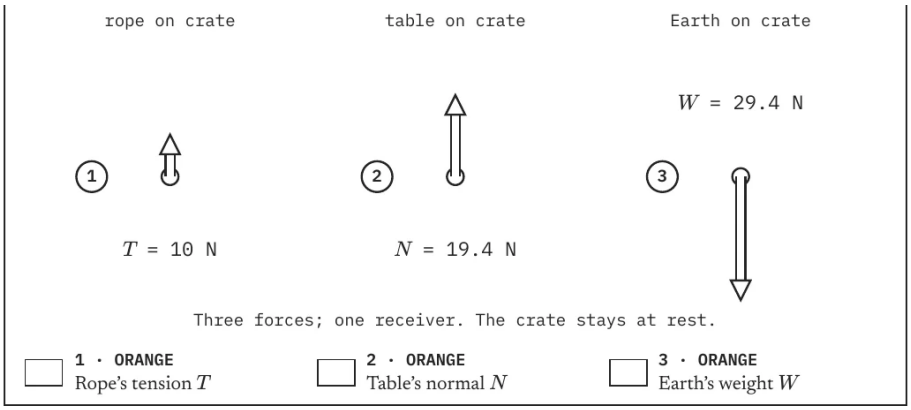
Also on your note page: Weight, Normal force, Tension.

B READ THE RELATIONSHIP

N + T − W = 0 for a crate at rest

As you read, find: source · receiver · contact.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 3 kg crate rests on a table. A vertical rope pulls up with 10 N . Find the table's normal force. Use g = 9.8 N/kg .

Given m = 3 kg T = 10 N up a y = 0

Rule W = mg . N + T − W = 0.

Work W = 3(9.8) = 29.4 N N = 29.4 − 10

Answer N = 19.4 N The rope shares the support.

E SAVE FOR THE FINAL MEETING

A forecast of XJ's changing motion needs real interactions, not a made-up motion force. Name the attracting body for each gravitational force before adding the forces.

Read the physics: Receiver

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Ask: Which object receives the force? What other object pushes or pulls it? A rope pulls by contact. Gravity pulls even without contact.

02 HOW IT WORKS

Mass is measured in kilograms. Weight is a force in newtons. On Earth, multiply mass by gravitational field strength to find weight. Earth pulls downward; a table pushes perpendicular to its surface.

03 WHAT TO CHECK

A rope may carry part of a crate's weight. If the crate stays at rest, all upward forces together balance the downward force.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 3 kg crate rests on a table. A vertical rope pulls up with 10 N . Find the table's normal force. Use g = 9.8 N/kg . N = 19.4 N The rope shares the support.

Model check: N + T − W = 0 for a crate at rest.

05 CONNECT TO XJ

A forecast of XJ's changing motion needs real interactions, not a made-up motion force. Name the attracting body for each gravitational force before adding the forces.

SAY IT BEFORE YOU WRITE

Who exerts the force, and on what object? The ___ exerts a force on the ___ toward ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Receiver / ES: objeto que recibe la fuerza / PT: objeto que recebe a força

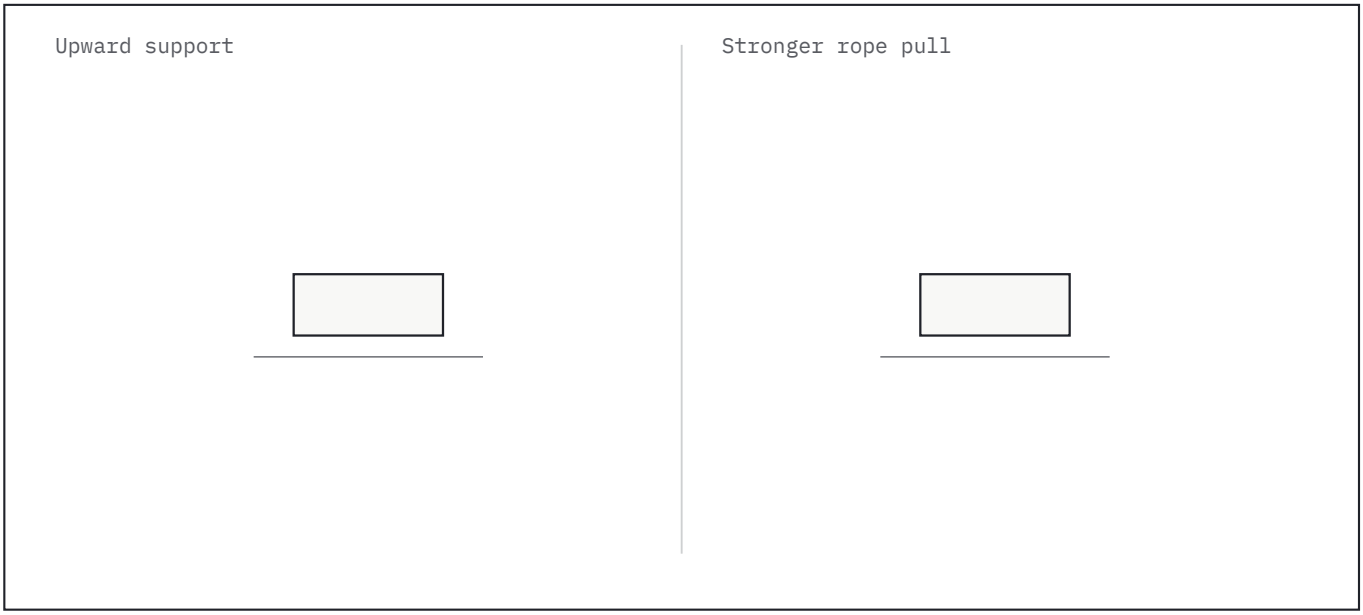
Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, receiver means …	Earth pulls on a crate. Which object feels that particular pull? The pull acts on ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Draw a supported crate before and after an upward rope pull. Use the arrow lengths to show how the table's force changes.

Labels you can use: source · receiver · contact.



Tell a partner: The ___ exerts a force on the ___ toward ___.

On assignment: Receiver

SECTION E / BACK / FINAL MEETING ONLY Name _____

An XJ animation shows gravity from Earth and a forward arrow labeled "because it moves." The graphics desk needs approved force labels.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Simulation engineer**

Prepare the force layer for XJ. Name a source and receiver for each arrow you keep. Put motion on a separate layer.

☐ **Science graphics editor**

Sketch the graphic viewers will see. Have your producer challenge the source of each arrow before you approve its label.

Word bank: source · receiver · contact.

Try: "Our desk recommends ___. This mark shows ___."

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

"We first said ___. Now we ___ because ___."

Editor's check: Identify another real source that would require its own force arrow.